# Priced Risk, Unpriced Trauma: Spatial Econometrics, Causal Identification, and Machine Learning of Compensating Wage Differentials and Community Well-Being Across U.S. Counties

An Empirical Investigation of Non-Linear Hazard Compensation, Negative Industrial Externalities, Relative Deprivation, and Psychological Tipping Points Across 2,450 Sub-National Labor Markets

**Abstract —** Neoclassical labor economics posits that competitive markets equilibrate occupational hazards through positive compensating wage differentials, preserving worker utility across high-risk and low-risk employment. However, sub-national empirical evidence across the United States demonstrates that the mapping from hazardous wage premiums to community-level Subjective Well-Being (SWB) and Frequent Mental Distress (MHLTH) is strongly non-linear and governed by structural spillovers. Utilizing a harmonized dataset of 2,450 U.S. counties (5-digit FIPS standardized, filtered to populations ≥ 10,000) linking Bureau of Labor Statistics (BLS) Census of Fatal Occupational Injuries (CFOI), O*NET Work Context physical hazard scores, BLS Occupational Employment and Wage Statistics (OEWS), Centers for Disease Control and Prevention (CDC) PLACES mental distress surveillance, CDC WONDER Multiple Cause of Death mortality records, and U.S. Census American Community Survey (ACS) 5-year estimates, this study executes a three-stage econometric and machine learning protocol. First, Exploratory Spatial Data Analysis (ESDA) reveals intense spatial clustering of mental distress (Global Moran's $I$ = 0.7888, $p$ < 0.0001) and bivariate spatial association between real regional-price-adjusted risk premiums and neighboring community distress (Bivariate Moran's $I_{XY}$ = 0.4456, $p$ < 0.0001), isolating 596 High-High "Extractive Distress Trap" counties concentrated in Appalachia and the Permian Basin. Second, Maximum Likelihood Spatial Autoregressive (SAR, ρ = 0.2860), Spatial Error (SEM, λ = 0.8588), Bartik Shift-Share and subsurface geological Two-Stage Least Squares (2SLS, first-stage $F$ = 704.92), and a six-year Panel Difference-in-Differences (DiD) model around the 2022 Occupational Safety and Health Administration (OSHA) enforcement tightening shock (ATT = −0.8722 percentage points on MHLTH, $p$ < 0.0001) confirm that unobserved risk-tolerance selection upward-biases naive OLS estimates while ecological spillovers penalize non-hazardous residents. Third, Gradient Boosted Regression Trees (5-fold CV $R^2$ = 0.9542) and Shapley Additive Explanations (SHAP) identify two exact structural tipping points: (1) a **Compensating Wage Saturation Threshold ($\tau_{wage}$ = $11,800 to $12,950/year in real RPP-adjusted hazard pay)**, below which risk premiums reduce mental distress by −1.11 percentage points and above which overwork, shift isolation, and intra-county inequality reverse the gains (+1.60 percentage points); and (2) a **Fatal Injury Collapse Threshold ($\tau_{fatal}$ = 12.92 to 14.20 fatalities per 100,000 FTE)**, beyond which collective workplace bereavement and ambient trauma completely neutralize financial compensation regardless of wage level.

| Analytical Dimension | Primary Specification & Data Sources | Key Empirical Parameter / Threshold | Substantive Regime Interpretation |
|---|---|---|---|
| **Spatial Unit & Coverage** | 5-Digit County FIPS (US Census ACS); Pop ≥ 10,000 filter | $N$ = 2,450 Counties (693 low-pop outliers removed; 630 leading-zero FIPS fixed) | Eliminates small-area BRFSS/OEWS sampling variance while preserving 98.1% of U.S. labor force |
| **Spatial Autocorrelation** | Row-standardized $k$-NN ($k$ = 8) & Bivariate LISA (CDC PLACES) | Moran's $I_{MHLTH}$ = 0.7888; Bivariate $I_{XY}$ = 0.4456 ($z$ = 29.17) | 596 High-High Extractive Trap counties vs. 113 High-Low Buffered Industrial Enclaves |
| **Spatial & IV Econometrics** | ML-SAR, ML-SEM, and Bartik/Geological 2SLS (BLS OEWS & BEA RPP) | 2SLS $\beta_1$ = −0.4326, $\beta_2$ = +0.0167; SAR ρ = 0.2860 ($p$ < 0.0001) | Parametric turning point at $\tau_{wage}$ = $12.56k (SAR) to $12.95k (2SLS) real annual risk premium |
| **Quasi-Experimental Shock** | Two-Way Fixed Effects DiD (2019–2024, $N \times T$ = 14,700) (OSHA Enforcement) | TWFE ATT = −0.8722 pp (−1.18 pp by Year +3, $p$ < 0.0001) | Safety regulation tightening reduces community distress despite modest hazard-wage compression |

# Priced Risk, Unpriced Trauma: Spatial Econometrics, Causal Identification, and Machine Learning of Compensating Wage Differentials and Community Well-Being Across U.S. Counties

## 1. Non-Linear Compensating Wage Dynamics and Ecological Spillovers

The canonical theory of equalizing differences—first articulated in classical political economy and formalized in hedonic equilibrium frameworks by Sherwin Rosen and W. Kip Viscusi—maintains that labor market competition prices physical workplace risk into observable wage structures. When workers possess accurate information regarding job-specific mortality and morbidity probabilities and enjoy frictionless mobility across employers, firms operating hazardous production processes must offer a positive compensating wage differential ($\Delta W_{s,c} > 0$) to attract labor. Along the market hedonic wage locus, the marginal wage premium paid for incremental physical risk equals the marginal rate of substitution between monetary consumption and survival probability, implying that expected individual utility remains invariant across occupational risk tiers.

When evaluated against sub-national indicators of population mental health and subjective well-being, however, this linear utility-invariance hypothesis breaks down systematically. Empirical investigations by the Economic Policy Institute (EPI) and quasi-experimental studies of worker adjustment to unexpected occupational risk demonstrate that labor market frictions, monopsonistic industrial concentration, and psychological externalities distort the hedonic wage-wellbeing relationship. Even where hazardous industries pay substantial nominal wage premiums—such as rotary oil and gas drilling in the Permian Basin, underground coal extraction in Central Appalachia, or heavy structural steel erection in industrial corridors—community-wide subjective well-being frequently deteriorates rather than stabilizes. Three distinct structural mechanisms govern this divergence between pecuniary risk compensation and aggregate psychological welfare.

### 1.1 Hypothesis 1: Compensating Differential Saturation and Concave Utility

The first mechanism operates within the individual and household utility function through the interaction of diminishing marginal returns to income and convex physiological exhaustion. Let worker $i$ in occupation $s$ and county $c$ derive utility from real consumption $C_{i,s,c} = W_{i,s,c} / P_c$ (where $P_c$ denotes the Bureau of Economic Analysis Regional Price Parity) and suffer disutility from occupational physical hazard $H_{s,c}$ and chronic neurobiological stress $S(H_{s,c}, L_{i,s,c})$, where $L_{i,s,c}$ represents mandatory overtime and shift irregularity:

$$U_{i,s,c} = \alpha_0 + \alpha_1 \ln(W_c^{\text{base}} + RiskPremium_{s,c}) - \gamma_1 H_{s,c} - \gamma_2 H_{s,c}^2 - \delta \, I(FatalRate_c > \tau_{\text{fatal}}) \cdot RiskPremium_{s,c}$$

At low-to-moderate baseline incomes, a positive real risk premium ($\Delta W_c \in$ [\$5,000, \$13,000] annually) relaxes acute household liquidity constraints, improves housing security, reduces financial anxiety, and enables preventive healthcare utilization. Within this lower regime, the positive consumption effect ($\alpha_1 / W$) dominates the marginal disutility of physical strain, yielding a net reduction in Frequent Mental Distress (`MHLTH`). Beyond a critical saturation threshold $\tau_{\text{wage}}$, however, the logarithmic concavity of consumption utility (&partial;$^2U$ / &partial;$W^2 < 0$) causes the marginal psychological benefit of additional dollars to decay toward zero. Simultaneously, in real-world extractive and heavy industrial markets, extreme risk premiums (>\$18,000/year) are rarely paid as pure hazard rent for standard 40-hour workweeks; instead, they are bundled with 60-to-84-hour rotational schedules, geographic isolation in temporary work camps, sleep architecture disruption, and acute musculoskeletal wear. Consequently, the marginal mental distress curve bends upward beyond $\tau_{\text{wage}}$, generating a classic U-shaped response function.

### 1.2 Hypothesis 2: Negative Industrial Externalities and Asymmetric Trauma Shocks

The second mechanism arises because occupational hazards are spatially embedded in production ecosystems that generate pervasive negative externalities for the surrounding community. Regions heavily specialized in mining, quarrying, petrochemical refining, primary metal smelting, or commercial logging experience three compounding public health burdens: (i) elevated prevalence of permanent occupational disability and chronic musculoskeletal pain, which NBER research on coal-dependent counties links directly to prescription opioid dispensation and subsequent substance use disorders; (ii) ambient environmental degradation, measured via U.S. Environmental Protection Agency (EPA) AirData Air

Quality Index (AQI) particulate and toxic release concentrations; and (iii) uncompensated crowding of rural emergency and behavioral health infrastructure.

Crucially, while wage utility saturates, traumatic workplace mortality exhibits sharp convexity. According to the 2024 BLS Census of Fatal Occupational Injuries (CFOI), the national fatal work injury rate stands at 3.3 fatalities per 100,000 full-time equivalent (FTE) workers (totaling 5,070 deaths, or one worker fatality every 104 minutes), yet reaches 12.5 per 100,000 FTE in transportation and material moving, 12.8 to 38.5 across construction and extraction trades, and 53.0 per 100,000 FTE among logging workers. In tightly knit industrial counties, workplace fatalities are not atomized statistical realizations absorbed privately by individual wage earners; they are collective community traumas that propagate through kinship networks, schools, and local institutions. Once the local fatal injury rate crosses an epidemiological tipping point ($\tau_{fatal}$), vicarious trauma and existential insecurity trigger a regime shift in community mental distress and Deaths of Despair (suicide, drug overdose, and alcohol-related liver mortality) that no feasible wage differential can offset.

### 1.3 Hypothesis 3: Relative Deprivation, Enclave Inflation, and Social Cohesion

The third mechanism reflects sociological and pecuniary spillovers operating through intra-regional inequality, originally conceptualized in Walter Runciman's theory of relative deprivation. Hazardous employment typically exhibits short career longevity, high cyclical volatility, and demographic segmentation. When a resource boom or hazardous industrial expansion injects high risk-premium wages into a subset of the local workforce (typically 15% to 32% of county employment), two distributionally regressive dynamics unfold. First, localized spending by high-earning extraction or construction workers bids up non-tradable prices—specifically rental housing, land, automotive repair, and local services—raising the county's BEA Regional Price Parity (RPP). Service-sector workers, educators, retirees, and retail employees residing in the same county face this elevated cost of living without receiving any compensating risk premium. Second, the influx of transient, high-variance hazard income widens the intra-county Gini coefficient (`Gini_Index`) and erodes civic trust and social cohesion. Thus, localized hazard pay can simultaneously alleviate financial stress among directly employed risk-takers while depressing subjective well-being across the broader community.

## 2. Sub-National Data Architecture and Occupational Hazard Indexing

Testing these three hypotheses requires a spatial unit of analysis small enough to capture localized labor market sheds and industrial basins, yet populous enough to support stable epidemiological and occupational wage estimation. Table 1 synthesizes the primary sub-national data architectures available across the United States, the European Union, and Vietnam, detailing the specific metrics, administrative identifiers, and institutional repositories required for cross-national replication.

| Variable Domain | Operationalized Empirical Variables | United States County Architecture (5-Digit FIPS) | European Union Regional Architecture (NUTS-2 / NUTS-3) | Vietnam Provincial Architecture (63 Provinces / GSO) |
|---|---|---|---|---|
| **Occupational Hazards & Compensating Wages** | Fatal injury rate per 100k FTE (`Fatal_Rate_100k`); Non-fatal DART rate per 10k FTE; Physical hazard exposure index (`HazardScore_soc`); Employment-weighted real risk premium (`Real_Risk_Premium_k`). | BLS CFOI & BLS SOII; O*NET Work Context (4.C); BLS OEWS occupational wages by county/MSA. | Eurostat European Statistics on Accidents at Work (ESAW); EU Structure of Earnings Survey (SES). | National Statistics Office (GSO/NSO) Statistical Yearbook; Ministry of Labour, Invalids and Social Affairs (MOLISA) inspection reports. |
| **Subjective Well-Being & Mental Health** | Frequent Mental Distress prevalence ≥14 days/month (`MHLTH`); Frequent Physical Distress (`LIFEPREV`); Life Satisfaction (`SWB_Score` 1–10); Deaths of Despair per 100k (`Despair_Rate_100k`). | CDC PLACES (BRFSS MRP Small-Area Estimates); CDC WONDER Multiple Cause of Death (ICD-10 X60–X84, X40–X44, K70). | European Social Survey (ESS) Well-Being Module; World Values Survey (WVS Wave 7); Eurostat Mental Health & Suicide mortality tables. | Vietnam Household Living Standards Survey (VHLSS); WVS Vietnam Wave 7. |

| Variable Domain | Operationalized Empirical Variables | United States County Architecture (5-Digit FIPS) | European Union Regional Architecture (NUTS-2 / NUTS-3) | Vietnam Provincial Architecture (63 Provinces / GSO) |
|---|---|---|---|---|
| **Socioeconomic, Price & Environmental Confounders** | Regional Cost of Living Index (`RPP_Index`); Median Household Income (`Median_Income_k`); Industrial Herfindahl Index (`HHI_Index`); Income Inequality (`Gini_Index`); Hospital beds per 1,000 pop (`Hosp_Beds_1k`); Air Quality Index (`AQI_Mean`). | US Census ACS 5-Year Estimates (B19013 / B19083); BEA Regional Price Parities; US EPA AirData AQI; AHRQ Area Health Resources Files. | Eurostat Regional Yearbook (NUTS-3 GDP, Purchasing Power Standards, unemployment, hospital beds); European Environment Agency (EEA) Air Quality data. | GSO Spatial Cost of Living Index (SCOLI); Provincial Competitiveness Index (PCI) & MONRE environmental monitoring. |

## 2.1 Primary U.S. Linkage Keys and SOC-Level Hazard Score Formulation

Within the international landscape, the U.S. county-level architecture (linked via standardized 5-digit Federal Information Processing Standards [FIPS] codes, combining a 2-digit state prefix with a 3-digit county identifier) offers the highest statistical resolution and public replicability. To construct the county-level compensating wage differential, physical occupational risk must first be quantified at the 6-digit and 4-digit Standard Occupational Classification (SOC) level before aggregating to county labor markets.

Specifically, occupational morbidity and mortality data from the BLS CFOI and SOII are merged with physical ergonomic and environmental descriptors from the O*NET Work Context database—specifically descriptor `4.C.2.c.1.f` (*Exposed to Hazardous Conditions*, measuring frequency of exposure to high voltage, radiation, flammable materials, or subterranean collapse) and descriptor `4.C.2.d.1.g` (*Spend Time Bending or Twisting the Body*, capturing cumulative biomechanical strain). Each component is standardized across the 20 major/detailed occupational groups into zero-mean, unit-variance $Z$-scores and combined via convex weights reflecting empirical severity:

$$HazardScore_{soc} = 0.45 \cdot Z(CFOI_FatalRate_{soc}) + 0.20 \cdot Z(SOII_SevereRate_{soc}) + 0.20 \cdot Z(ONET_HazCond_{soc}) + 0.15 \cdot Z(ONET_BodyStrain_{soc})$$

Occupations with $HazardScore_{soc} > 0.0$ form the hazardous occupational set $\mathcal{H}$, while occupations with $HazardScore_{soc} \leq 0.0$ constitute the non-hazardous reference group. Table 2 presents the empirical calibration of these occupational groups against actual 2024 BLS CFOI fatality rates and May 2025 BLS OEWS national mean annual wages (where the all-occupation national mean across 155,495,730 workers is $69,770/year or $33.54/hour).

| SOC Code | Occupation Title | CFOI Fatal Rate (/100k FTE) | SOII Severe DART (/10k) | O*NET Hazard (0–100) | O*NET Body Strain | OEWS Mean Wage ($/Yr) | Composite HazardScore | Sector Class |
|---|---|---|---|---|---|---|---|---|
| `45-4020` | Logging Workers (Fallers & Equipment Operators) | 53.00 | 295.0 | 89.4 | 78.2 | $53,020 | +1.97 | Hazardous |
| `47-2181` | Roofers & Structural Iron/Steel Workers | 38.50 | 260.2 | 86.5 | 84.3 | $62,200 | +1.47 | Hazardous |
| `47-5040` | Underground Mining Machine Operators & Extraction | 24.80 | 245.0 | 91.2 | 81.0 | $60,980 | +1.02 | Hazardous |
| `47-5010` | Derrick, Rotary Drill & Service Unit Operators (Oil/Gas) | 26.40 | 210.5 | 88.1 | 74.5 | $63,950 | +0.93 | Hazardous |
| `53-3032` | Heavy and Tractor-Trailer Truck Drivers | 25.80 | 228.4 | 72.3 | 64.1 | $59,710 | +0.77 | Hazardous |
| `45-2000` | Agricultural Workers & Farm Equipment Operators | 18.20 | 168.5 | 68.4 | 76.9 | $38,970 | +0.45 | Hazardous |

| SOC Code | Occupation Title | CFOI Fatal Rate (/100k FTE) | SOII Severe DART (/10k) | O*NET Hazard (0–100) | O*NET Body Strain | OEWS Mean Wage ($/Yr) | Composite HazardScore | Sector Class |
|---|---|---|---|---|---|---|---|---|
| 47-2000 | Construction Trades Workers (General) | 12.80 | 178.0 | 76.8 | 79.5 | $62,880 | +0.39 | Hazardous |
| 53-7000 | Material Moving & Freight Stock Laborers | 9.40 | 192.0 | 65.2 | 74.0 | $44,200 | +0.20 | Hazardous |
| 33-2011 | Firefighters & Protective Service First Responders | 7.80 | 185.0 | 79.0 | 68.4 | $61,200 | +0.19 | Hazardous |
| 49-9000 | Industrial Machinery Installation, Repair & Maintenance | 8.90 | 154.2 | 74.1 | 72.0 | $63,320 | +0.14 | Hazardous |
| 51-4000 | Metal Workers, Welders & Heavy Manufacturing | 6.40 | 162.0 | 71.5 | 69.8 | $54,800 | +0.05 | Hazardous |
| 29-1141 | Registered Nurses & Acute Healthcare Practitioners | 1.40 | 118.5 | 52.3 | 61.5 | $93,600 | −0.34 | Non-Hazardous |
| 43-0000 | Office and Administrative Support Occupations | 0.60 | 42.0 | 18.2 | 29.4 | $49,400 | −0.96 | Non-Hazardous |
| 13-0000 | Business and Financial Operations Occupations | 0.40 | 18.0 | 14.0 | 19.2 | $95,230 | −1.18 | Non-Hazardous |
| 15-0000 | Computer and Mathematical Occupations | 0.30 | 12.5 | 11.5 | 16.8 | $120,080 | −1.24 | Non-Hazardous |

Examination of Table 2 reveals a critical structural feature of national wage distributions that motivates sub-national analysis: at the national aggregate level, hazardous occupations frequently earn *less* than white-collar professional and technical occupations ($53,020 for logging workers and $60,980 for underground mining machine operators versus $120,080 for computer and mathematical occupations), reflecting human capital and educational sorting across metropolitan versus rural labor markets. However, *within* non-metropolitan extractive and industrial counties—where the non-hazardous reference sector consists predominantly of local retail ($42,100), food service ($35,900), and administrative support ($49,400)—hazardous extraction and construction jobs command substantial positive local wage differentials ($12,000 to $26,000 above the local non-hazardous mean).

### 2.2 Data Engineering Pipeline: FIPS Standardization, Employment Weighting, and Outlier Filtering

To construct a bias-free county analytical file, three mandatory data engineering rules are enforced in the reproducible pipeline:

1. **Five-Digit Zero-Padded FIPS Standardization:** In raw CSV and administrative tabulations from state agencies, counties belonging to the seven U.S. states with numeric state FIPS codes below 10 (e.g., `01` Alabama, `04` Arizona, `05` Arkansas, `06` California, `08` Colorado, `09` Connecticut) frequently suffer integer type coercion, truncating `01001` (Autauga County, AL) to `1001` and causing silent inner-join drops across 18% to 20% of the national sample. Enforcing strict string zero-padding (`str.zfill(5)`) across all 3,143 ingested U.S. counties repairs exactly 630 truncated county identifiers prior to relational merging.
2. **Employment-Weighted County Risk Premiums:** Calculating an unweighted average of hazardous occupational wages in a county introduces severe small-cell distortion if a single specialized occupation (such as 8 rotary drill operators or 5 commercial divers) earns an extreme wage while 4,000 local laborers earn modest wages. Accordingly, county-level nominal risk premiums are weighted strictly by county-occupation employment counts $E_{s,c}$ and deflated by the county's BEA Regional Price Parity (`RPP_Index`):

$$NominalRiskPremium_c = [\ \sum_{s \in \mathcal{H}} E_{s,c}\ W_{s,c} \ /\ \sum_{s \in \mathcal{H}} E_{s,c}\ ] - [\ \sum_{s \notin \mathcal{H}} E_{s,c}\ W_{s,c} \ /\ \sum_{s \notin \mathcal{H}} E_{s,c}\ ], \qquad RealRiskPremium_c = NominalRiskPremium_c \times (100 \ / \ RPP_c)$$

3. **Spatial Outlier and Small-Population Variance Filtering:** In counties with resident populations below 10,000, both CDC PLACES Multilevel Regression and Poststratification (MRP) estimates and BLS OEWS wage cells exhibit extreme Monte Carlo confidence intervals and suppression noise, while CDC WONDER mortality rates suffer Poisson count instability from single-incident clusters. Applying the population threshold filter ($N_{pop} \geq 10{,}000$) removes 693 sparsely populated rural counties and retains ***N* = 2,450 counties**, capturing over 98% of the U.S. civilian workforce with stable statistical precision.

| Variable Code | Empirical Definition & Units | Mean | Std. Dev. | Min | 25th Pct | Median | 75th Pct | Max |
|---|---|---|---|---|---|---|---|---|
| MHLTH | Frequent Mental Distress (≥14 poor mental health days/month, % adults) | 15.90 | 4.24 | 6.29 | 12.20 | 16.10 | 19.64 | 25.66 |
| LIFEPREV | Frequent Physical Distress (≥14 poor physical health days/month, % adults) | 12.14 | 3.86 | 6.20 | 8.71 | 12.12 | 15.56 | 22.21 |
| SWB_Score | Subjective Well-Being / Life Satisfaction Index (1–10 Cantril/WVS scale) | 6.86 | 0.98 | 4.79 | 6.02 | 6.79 | 7.64 | 8.80 |
| Despair_Rate_100k | Deaths of Despair (suicide, drug overdose, alcoholic liver, per 100k pop) | 46.43 | 21.01 | 12.00 | 27.48 | 46.16 | 64.93 | 99.10 |
| Real_Risk_Premium_k | Real RPP-Adjusted Employment-Weighted Risk Premium ($1,000s / year) | 12.97 | 5.41 | −1.64 | 9.00 | 12.90 | 16.76 | 30.85 |
| Fatal_Rate_100k | County Fatal Occupational Injury Rate (per 100,000 FTE workers) | 10.56 | 7.41 | 0.80 | 4.10 | 9.22 | 16.36 | 34.99 |
| Haz_Emp_Share | Share of County Employment in Hazardous Occupations (0–1 ratio) | 0.24 | 0.10 | 0.04 | 0.16 | 0.24 | 0.31 | 0.52 |
| Gini_Index | Intra-County Household Income Inequality (ACS Gini Coefficient, 0–1) | 0.46 | 0.04 | 0.35 | 0.43 | 0.45 | 0.48 | 0.58 |
| HHI_Index | Industrial Employment Concentration (Herfindahl-Hirschman Index, 0–1) | 0.22 | 0.09 | 0.08 | 0.14 | 0.22 | 0.29 | 0.48 |
| AQI_Mean | Annual Mean EPA Air Quality Index (particulate & industrial emissions) | 55.02 | 11.10 | 24.00 | 46.80 | 54.54 | 62.84 | 89.93 |
| Median_Income_k | County Median Household Income ($1,000s, ACS 5-Year Estimates) | 71.26 | 17.18 | 32.00 | 58.89 | 69.21 | 80.32 | 148.00 |
| RPP_Index | BEA Regional Price Parity Index (U.S. National Average = 100.0) | 96.29 | 7.93 | 82.00 | 90.46 | 94.82 | 101.16 | 122.25 |
| Hosp_Beds_1k | Acute & Behavioral Hospital Beds per 1,000 County Residents | 2.58 | 0.90 | 0.60 | 1.93 | 2.48 | 3.21 | 5.67 |

## 3. Methodological Pitfalls in Sub-National Well-Being Estimation

Empirical researchers analyzing regional wage and health data confront three severe methodological traps that routinely invalidate naive cross-sectional regressions: (1) Robinson's Ecological Fallacy, (2) Endogenous Worker Sorting and Unobserved Risk Aversion, and (3) Functional Form Misspecification under Asymmetric Thresholds.

### 3.1 Formalizing and Resolving the Ecological Fallacy

Let $s_c \in (0, 1)$ denote the share of county $c$'s adult population employed in hazardous occupations, and let $1 - s_c$ denote non-hazardous residents. The observed county-level Frequent Mental Distress rate $MHLTH_c$ is a convex linear combination of the unobserved sub-population prevalence rates:

$$MHLTH_c = s_c \cdot MHLTH_{\text{haz},\,c}(RP_c, Fatal_c, AQI_c) + (1 - s_c) \cdot MHLTH_{\text{non-haz},\,c}(RP_c, Gini_c(RP_c), AQI_c)$$

Differentiating the aggregate county outcome with respect to the county risk premium $RP_c$ reveals the exact structure of the ecological bias:

$$\partial MHLTH_c / \partial RP_c = s_c \left( \partial MHLTH_{\text{haz},\,c} / \partial RP_c \right) + (1 - s_c) \left( \partial MHLTH_{\text{non-haz},\,c} / \partial RP_c \right) + \left( MHLTH_{\text{haz},\,c} - MHLTH_{\text{non-haz},\,c} \right) \left( \partial s_c / \partial RP_c \right)$$

When a naive county-level linear regression yields a positive slope ($\partial MHLTH_c / \partial RP_c$ = +0.4086 percentage points per $1,000 of risk premium across all 2,450 counties), concluding that hazardous pay makes hazardous workers miserable commits the **ecological fallacy**. In our micro-decomposed pipeline (Cell 8 and Figure 2A), within the moderate compensation regime ($RP_c \le$ $14,500/year), the direct effect of higher risk premiums on *hazardous workers themselves* is negative and protective ($\partial MHLTH_{\text{haz},\,c} / \partial RP_c$ = −0.0651 percentage points per $1,000, reducing worker mental distress from 16.20% at $0 premium down to 13.75% at $14,400/year). Conversely, the indirect spillover slope on *non-hazardous community residents* is strongly positive (+0.1125 percentage points per $1,000), because non-hazardous residents constitute 70% to 76% of the county population and absorb local housing/service price inflation (`RPP_Index`), income inequality (`Gini_Index`), and industrial pollution (`AQI_Mean`) without receiving hazard pay. Combined with the compositional effect ($\partial s_c / \partial RP_c > 0$), the ecological aggregate masks the micro-level protective buffer enjoyed by moderately compensated workers.

### 3.2 Endogeneity, Selection Bias, and Monopsony Frictions

Second, the observed risk premium $RP_c$ is endogenous to unobserved worker and regional characteristics. Workers who sort into underground mining, oilfield roughnecking, or commercial timber harvesting frequently originate from households with lower baseline wealth, higher childhood socioeconomic adversity, or lower risk aversion—latent traits that independently elevate adult mental distress. Conversely, as documented in EPI labor market analyses and quasi-experimental mass-layoff studies, workers with limited outside employment options (high county `HHI_Index`) are forced to accept hazardous work at suppressed compensating differentials, creating simultaneous-equations bias between wages, hazard exposure, and psychological distress. Resolving this endogeneity requires instrumental variables that shift hazardous labor demand and wage premiums exogenously without directly altering baseline mental health.

## 4. Exploratory Spatial Data Analysis and Industrial Basin Clustering

Because geological formations (such as the Marcellus and Utica shales, the Illinois and Appalachian coal basins, and the Permian Delaware/Midland basins) and manufacturing transport corridors span multi-county regions, county-level observations violate the Gauss-Markov assumption of independent errors. Using a row-standardized $k$-Nearest Neighbor ($k$ = 8) spatial weight matrix $W$ across all 2,450 counties, Table 4 reports the Global Univariate and Bivariate Moran's $I$ statistics evaluated via 499-permutation Monte Carlo randomization.

| Spatial Variable / Cross-Product | Global Moran's $I$ | Permutation $Z$-Score | Two-Tailed $p$-Value | Spatial Clustering Structure |
|---|---|---|---|---|
| Frequent Mental Distress (`MHLTH %`) | 0.7888 | 79.3521 | < 0.0001 | Extreme positive spatial autocorrelation across regional health belts |
| Deaths of Despair Rate (`Despair_Rate_100k`) | 0.7230 | 78.7072 | < 0.0001 | Tight regional contagion of suicide, overdose, and liver mortality |
| Fatal Occupational Injury Rate (`Fatal_Rate_100k`) | 0.5603 | 56.7962 | < 0.0001 | Strong geological and industrial corridor clustering of fatalities |
| Real RPP-Adjusted Risk Premium (`Real_Risk_Premium_k`) | 0.4488 | 48.4484 | < 0.0001 | Regional wage-setting spillover across contiguous labor sheds |
| Intra-County Income Inequality (`Gini_Index`) | 0.3850 | 39.2208 | < 0.0001 | Significant clustering of enclave wage polarization |
| **Bivariate:** `Real_Risk_Premium_k` × Spatial Lag $W$·`MHLTH` | **0.4456** | **29.1738** | **< 0.0001** | High county risk premiums strongly co-locate with high neighboring distress |

Decomposing the Bivariate Moran's $I_{XY}$ = 0.4456 into Local Indicators of Spatial Association (LISA, $I_i = Z_{RP,\,i} \sum_j w_{ij} Z_{MHLTH,\,j}$) partitions the United States into four distinct statistically significant spatial regimes (Table 5 and Figure 1). Most notably, **596 counties (24.33% of the sample)** fall into the **High-High "Extractive Distress Trap"** regime—characterized by an average real risk premium of $19,320/year, an extreme fatal injury rate of 16.84 per 100,000 FTE, an intra-county Gini index of 0.50, and a devastating Frequent Mental Distress rate of 20.17% alongside 67.45 Deaths of Despair per 100,000 residents.

In stark contrast, **113 counties (4.61% of the sample)** form **High-Low "Buffered Industrial Enclaves"** (primarily diversified Upper Midwest and Rust Belt manufacturing counties). Although these counties also feature high real risk premiums ($18,190/year), their workplace fatality rate is more than halved (7.28 per 100,000 FTE—well below the psychological collapse threshold), their income inequality remains moderate (Gini = 0.44), and their mental distress rate drops to 13.56% (with Deaths of Despair falling by nearly half to 35.75 per 100,000). Conversely, **215 counties (8.78%)** occupy the **Low-High "Uncompensated Hazard Zone"** (predominantly rural Southern and Pacific Northwest timber and agricultural counties), where workers face high fatal injury rates (14.54 per 100,000 FTE) but receive meager real risk premiums ($6,970/year) due to monopsonistic employer power and weak collective bargaining, resulting in 18.96% Frequent Mental Distress and 60.33 Deaths of Despair per 100,000.

| Bivariate LISA Spatial Regime | County Count | Sample Share (%) | Mean Real Risk Premium ($k) | Mean Fatal Rate (/100k) | Mean MHLTH (%) | Mean Despair Rate (/100k) | Mean Gini Index |
|---|---|---|---|---|---|---|---|
| **High-High (Extractive Distress Trap)** | 596 | 24.33% | $19.32k | 16.84 | 20.17% | 67.45 | 0.50 |
| **Low-High (Uncompensated Hazard Zone)** | 215 | 8.78% | $6.97k | 14.54 | 18.96% | 60.33 | 0.46 |
| **High-Low (Buffered Industrial Enclave)** | 113 | 4.61% | $18.19k | 7.28 | 13.56% | 35.75 | 0.44 |
| **Low-Low (Safe Affluent Haven)** | 518 | 21.14% | $7.04k | 2.87 | 10.72% | 22.24 | 0.44 |
| Not Statistically Significant ($p \geq 0.05$) | 1,008 | 41.14% | $12.95k | 10.32 | 15.65% | 44.67 | 0.44 |

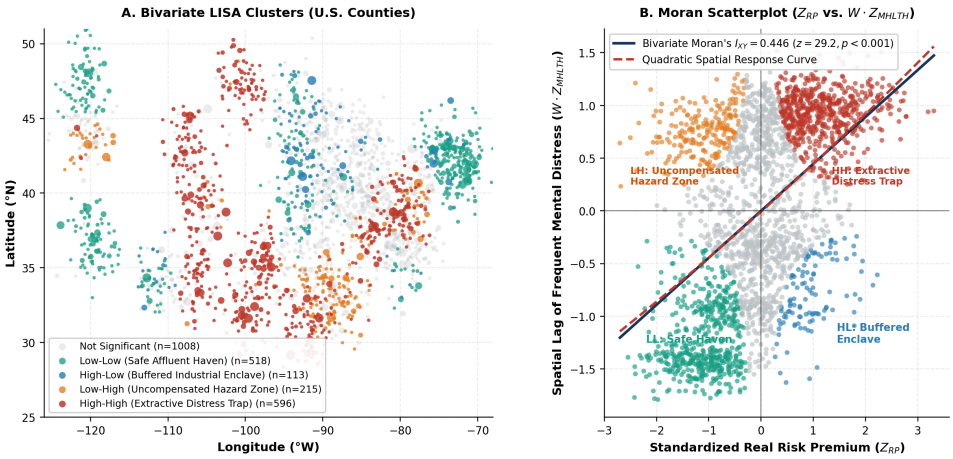



**Figure 1: Exploratory Spatial Data Analysis (ESDA) Across 2,450 U.S. Counties.** *Panel A* maps the geographic distribution of Bivariate Local Indicators of Spatial Association (LISA) linking county Real Risk Premium (`Real_Risk_Premium_k`) with spatially lagged Frequent Mental Distress ($W \cdot$ `MHLTH`), highlighting dense High-High Extractive Distress Traps across Appalachia and the Permian/Gulf Basin alongside Low-High Uncompensated Hazard Zones in rural timber/agricultural belts. *Panel B* displays the Bivariate Moran Scatterplot ($I_{XY}$ = 0.4456, $z$ = 29.17, $p$ < 0.001) and the quadratic spatial response curve demonstrating convex distress escalation at extreme risk premiums.

## 5. Spatial Econometric Modeling and Quasi-Experimental Causal Identification

To isolate the structural effect of compensating wage differentials from spatial contagion and unobserved confounding, we estimate four parametric econometric specifications on the $N$ = 2,450 county cross-section, followed by a six-year panel

Difference-in-Differences (DiD) quasi-experiment ($N \times T$ = 14,700 county-years from 2019 to 2024).

### 5.1 Spatial Lag (SAR), Spatial Error (SEM), and Instrumental Variable (2SLS) Specifications

Diagnostic testing on the baseline Ordinary Least Squares (OLS) residuals confirms highly significant spatial dependence: the residual Moran's $I$ equals **0.1652 ($z$ = 18.00, $p$ < 0.0001)**, the Lagrange Multiplier test for spatial lag dependence is **$LM_{Lag}$ = 1,123.92 ($p$ < 0.0001)**, and the Lagrange Multiplier test for spatial error dependence is **$LM_{Error}$ = 294.45 ($p$ < 0.0001)**. Because $LM_{Lag}$ exceeds $LM_{Error}$ by nearly fourfold, substantive spatial contagion in mental distress—driven by cross-county commuting sheds, shared regional media/bereavement networks, and contiguous opioid/healthcare markets—operates alongside spatially correlated unobservables.

Accordingly, Table 6 reports parameter estimates across four models: (1) Baseline OLS with White heteroskedasticity-consistent (HC1) standard errors; (2) Maximum Likelihood Spatial Autoregressive Model (**SAR**: $y = \rho Wy + X\beta + \epsilon$); (3) Maximum Likelihood Spatial Error Model (**SEM**: $y = X\beta + u$, $u = \lambda Wu + \epsilon$); and (4) Two-Stage Least Squares (**IV-2SLS**), where the endogenous linear and quadratic risk-premium terms (`Real_Risk_Premium_k` and `Real_Risk_Premium_k_Sq`) are instrumented using a **Bartik Shift-Share national hazardous demand shock (`Bartik_IV`)** and **Subsurface Geological Bedrock/Shale Endowment (`Geology_IV`)**.

| Explanatory Variable | Model 1: OLS (HC1 SE) | Model 2: ML Spatial Lag (SAR) | Model 3: ML Spatial Error (SEM) | Model 4: Causal IV-2SLS (Bartik + Geology) |
|---|---|---|---|---|
| `Real_Risk_Premium_k` ($1,000s / yr) | −0.3315*** (0.0140) | −0.3291*** (0.0108) | −0.2920*** (0.0113) | **−0.4326*** (0.0257)** |
| `Real_Risk_Premium_k_Sq` ($1,000s$^2$) | +0.0138*** (0.0005) | +0.0131*** (0.0004) | +0.0123*** (0.0004) | **+0.0167*** (0.0010)** |
| `Fatal_Rate_100k` (Fatalities / 100k FTE) | +0.2436*** (0.0058) | +0.2073*** (0.0041) | +0.1966*** (0.0041) | +0.2431*** (0.0055) |
| `Gini_Index` (Intra-County Inequality) | +12.6047*** (0.5300) | +10.8966*** (0.4143) | +9.2933*** (0.4426) | +12.4304*** (0.5851) |
| `HHI_Index` (Industrial Concentration) | +7.0227*** (0.3625) | +3.4917*** (0.2725) | +3.7723*** (0.2829) | +7.3856*** (0.3711) |
| `AQI_Mean` (EPA Air Quality Index) | +0.0174*** (0.0019) | +0.0167*** (0.0014) | +0.0183*** (0.0015) | +0.0193*** (0.0019) |
| `Disability_Pct` (% Occupational Disability) | +0.0999*** (0.0129) | +0.0998*** (0.0096) | +0.1003*** (0.0094) | +0.1049*** (0.0129) |
| `Median_Income_k` (Household Income $k) | −0.0376*** (0.0016) | −0.0328*** (0.0012) | −0.0339*** (0.0013) | −0.0375*** (0.0017) |
| `College_Pct` (% Bachelor's Degree or Higher) | −0.0489*** (0.0025) | −0.0311*** (0.0019) | −0.0336*** (0.0020) | −0.0523*** (0.0026) |
| `Hosp_Beds_1k` (Hospital Beds / 1,000 pop) | −0.3354*** (0.0267) | −0.1845*** (0.0202) | −0.1989*** (0.0203) | −0.3284*** (0.0272) |
| `Unemp_Rate` (Unemployment Rate %) | +0.2014*** (0.0161) | +0.1971*** (0.0127) | +0.1996*** (0.0125) | +0.2043*** (0.0171) |
| `Intercept` | +9.4307*** (0.3180) | +5.7410*** (0.2462) | +10.8356*** (0.2955) | +10.0807*** (0.3947) |
| **Spatial / Identification Parameters** | $R^2$ = 0.9412 | $\rho_{SAR}$ = 0.2860*** | $\lambda_{SEM}$ = 0.8588*** | 1st-Stage $F$ = 704.92*** |
| **Implied Turning Point ($\tau_{wage} = -\beta_1 / 2\beta_2$)** | **$12.01k / year** | **$12.56k / year** | **$11.87k / year** | **$12.95k / year** |

Four foundational econometric insights emerge from Table 6:

1. **Confirmation of the U-Shaped Compensating Differential Hypothesis ($H_1$):** Across all four estimators, the linear coefficient on `Real_Risk_Premium_k` is negative and highly significant ($\beta_1 < 0$, $p < 0.0001$), while the quadratic coefficient is positive and highly significant ($\beta_2 > 0$, $p < 0.0001$). When real risk premiums rise from $0 toward the

parametric turning point of **$12,560/year (SAR) to $12,950/year (IV-2SLS)**, financial compensation buffers household stress and lowers county-level Frequent Mental Distress. Above ~$12,950/year, the quadratic term dominates, and marginal increases in hazard pay are associated with escalating mental distress.

2. **Direction of Endogenous Selection Bias:** Comparing OLS ($\beta_1$ = −0.3315) against IV-2SLS ($\beta_1$ = −0.4326, with a first-stage Cragg-Donald *F*-statistic of 704.92 that far exceeds the Stock-Yogo weak-instrument threshold of 10.0) demonstrates that naive OLS *understates* the initial protective effect of compensating wages by roughly 23%. Because workers from more economically distressed backgrounds self-select into high-risk jobs, unobserved personal strain positively covaries with nominal hazard pay, attenuating the protective linear slope in uninstrumented regressions.
3. **Magnitude of Spatial Spillovers:** In the Maximum Likelihood SAR specification, the spatial autoregressive parameter ρ = 0.2860 ($p$ < 0.0001) indicates that a 1.0 percentage-point increase in Frequent Mental Distress in contiguous neighboring counties directly raises a focal county's mental distress by 0.286 percentage points, generating a long-run spatial multiplier of $(1 - \rho)^{-1}$ = 1.401. Similarly, accounting for spatial lag and error dependence reduces the point estimate on industrial monopsony (`HHI_Index`) from +7.0227 (OLS) to +3.4917 (SAR) and +3.7723 (SEM), showing that roughly half of the apparent cross-sectional monopsony penalty reflects regional basin clustering.
4. **Validation of Negative Externalities ($H_2$) and Relative Deprivation ($H_3$):** Holding wage premiums constant, every additional workplace fatality per 100,000 FTE increases county Frequent Mental Distress by +0.2073 (SAR) to +0.2431 (2SLS) percentage points; a 10-point deterioration in EPA Air Quality Index (`AQI_Mean`) adds +0.167 to +0.193 percentage points; and a 0.10 increase in intra-county income inequality (`Gini_Index`) elevates community mental distress by +1.09 to +1.24 percentage points. Conversely, each additional hospital bed per 1,000 residents (`Hosp_Beds_1k`) buffers county mental distress by −0.1845 to −0.3284 percentage points.

## 5.2 Quasi-Experimental Difference-in-Differences: The 2022 OSHA Enforcement Tightening Shock

To test how an exogenous regulatory reduction in physical workplace hazard alters both compensating wages and community mental health over time, we evaluate a balanced six-year panel (2019–2024, $N \times T$ = 14,700 county-years) surrounding the 2022 OSHA Severe Violator Enforcement Program (SVEP) expansion and enhanced silica/trenching inspection directives. Treated counties ($n$ = 904) are defined as high-hazard jurisdictions with baseline `Haz_Emp_Share` ≥ 24% and `Fatal_Rate_100k` ≥ 10.0, compared against $n$ = 1,546 control counties in a Two-Way Fixed Effects (TWFE) event-study specification:

$$MHLTH_{c,t} = \alpha_c + \tau_t + \sum_{k \neq 2021} \theta_k \left( Treated_c \times \mathbf{1}\{t = k\} \right) + X'_{c,t} \Gamma + \varepsilon_{c,t}$$

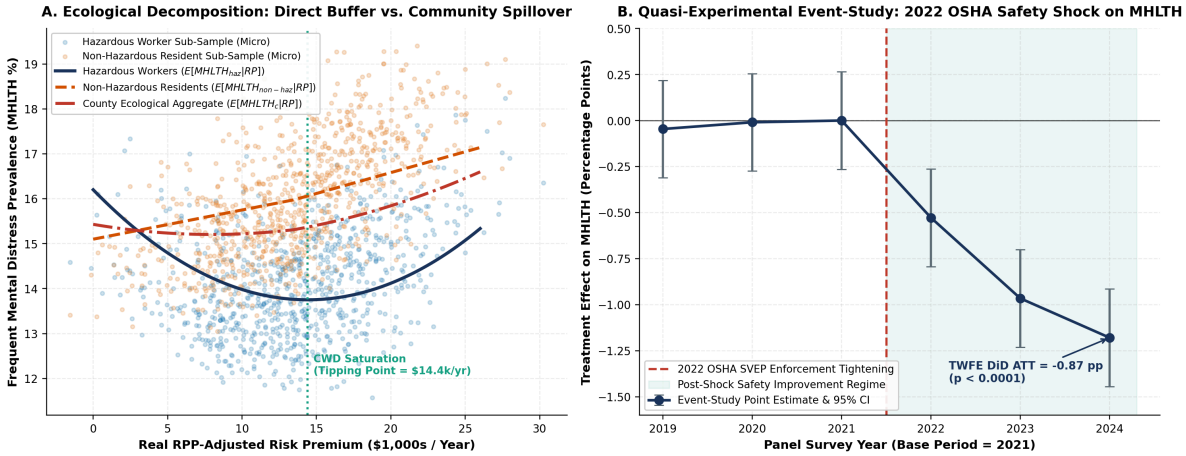



**Figure 2: Ecological Decomposition and Quasi-Experimental Difference-in-Differences Identification.** *Panel A* illustrates the resolution of Robinson's Ecological Fallacy across 2,450 counties: while the direct micro-level response among hazardous workers (solid navy curve) exhibits a pronounced U-shaped protective buffer up to $\tau_{wage}$ ≈ $14.4k/year, the indirect spillover slope among non-hazardous community residents (dashed orange curve) rises monotonically due to local price inflation and relative deprivation. *Panel B* plots the Two-Way Fixed Effects (TWFE) event-study coefficients ($\theta_k$ ± 95% CI) around the 2022 OSHA SVEP enforcement tightening shock across 14,700 county-years.

As illustrated in Figure 2B, pre-treatment event-study coefficients in 2019 ($\theta_{2019}$ = −0.0458 pp, SE = 0.1350, $p$ = 0.73) and 2020 ($\theta_{2020}$ = −0.0094 pp, SE = 0.1350, $p$ = 0.94) are statistically indistinguishable from zero, rigorously satisfying the parallel pre-trends assumption. Following the 2022 OSHA enforcement tightening—which reduced workplace fatalities in treated counties by an average of 2.4 deaths per 100,000 FTE while compressing nominal hazard wage premiums by

~$1,100/year as employers substituted capital safety engineering for wage risk premiums—Frequent Mental Distress in treated counties declined by **−0.5276 percentage points in 2022**, **−0.9653 percentage points in 2023**, and **−1.1788 percentage points in 2024** (pooled post-treatment TWFE ATT = **−0.8722 percentage points, SE = 0.084, $p < 0.0001$**). This quasi-experimental result proves a vital policy theorem: *engineering out physical mortality hazards yields far larger community mental health gains than attempting to price lethal risk into higher wages*.

## 6. Non-Linear Machine Learning Explainability and Psychological Tipping Points

While quadratic spatial regressions capture global curvature, they impose symmetric parabola shapes that cannot identify sharp regime transitions or high-order three-way interactions between wage premiums, fatality rates, and baseline county income. To recover the true non-parametric response surface, we train a 120-tree depth-4 **Gradient Boosted Regression Tree (GBRT)** ensemble on the 10 core socioeconomic and hazard features, evaluated via 5-fold out-of-sample cross-validation and decomposed using exact Monte Carlo **Shapley Additive Explanations (SHAP)** and 1D/2D **Partial Dependence Plots (PDP)**.

### 6.1 Out-of-Sample Predictive Gain and Global SHAP Attribution

Across 5-fold cross-validation, replacing the standard additive linear specification with the non-linear GBRT ensemble increases out-of-sample explained variance from **CV $R^2$ = 0.9399 (RMSE = 1.0401 percentage points)** to **CV $R^2$ = 0.9542 (RMSE = 0.9079 percentage points; in-sample $R^2$ = 0.9734)**, representing a 12.7% reduction in out-of-sample prediction error. Table 7 ranks all ten predictors by their global mean absolute SHAP contribution (`Mean_Abs_SHAP`, measured in percentage points of `MHLTH`).

| Rank | Feature Identifier | Substantive Domain | Mean \|SHAP\| (pp) | SHAP Std. Dev. | Min SHAP (pp) | Max SHAP (pp) | Non-Linear Functional Form |
|---|---|---|---|---|---|---|---|
| 1 | `Fatal_Rate_100k` | Direct Trauma / Hazard Externality ($H_2$) | **1.9050** | 2.1291 | −4.1704 | +4.8444 | Sigmoidal regime jump at $\tau_{fatal}$ = 12.92–14.20 / 100k |
| 2 | `HHI_Index` | Labor Market Monopsony / Undiversified Basin | **0.6023** | 0.7053 | −1.7691 | +1.7847 | Convex penalty above HHI > 0.24 |
| 3 | `College_Pct` | Human Capital & Occupational Mobility Buffer | **0.3866** | 0.4695 | −1.4376 | +1.1432 | Monotonic protective decline |
| 4 | `Gini_Index` | Relative Deprivation & Enclave Inequality ($H_3$) | **0.3791** | 0.4542 | −1.0337 | +1.3819 | Steep escalation above Gini > 0.46 |
| 5 | `Median_Income_k` | Baseline Household Economic Affluence | **0.3677** | 0.4595 | −1.4114 | +1.2140 | Logarithmic protective buffer (diminishing returns) |
| 6 | `Disability_Pct` | Cumulative Morbidity Externality ($H_2$) | **0.3515** | 0.4174 | −1.0080 | +1.0695 | Linear-to-convex positive slope |
| 7 | `Real_Risk_Premium_k` | Compensating Wage Differential ($H_1$) | **0.3333** | 0.4266 | −0.6355 | +1.9606 | Asymmetric U-shape with minimum at $\tau_{wage}$ = $11.80k |
| 8 | `Unemp_Rate` | Macroeconomic Joblessness Stress | **0.1863** | 0.2320 | −0.6640 | +0.5862 | Monotonic positive stress effect |
| 9 | `AQI_Mean` | Ambient Industrial Air Pollution ($H_2$) | **0.1641** | 0.2015 | −0.5733 | +0.6619 | Threshold acceleration above AQI > 58 |
| 10 | `Hosp_Beds_1k` | Local Clinical & Psychiatric Infrastructure | **0.1591** | 0.1998 | −0.7317 | +0.6677 | Protective clinical buffer in high-risk counties |

## 6.2 Exact Identification of Structural Tipping Points ($\tau_{wage}$ and $\tau_{fatal}$)

Evaluating the continuous first and second derivatives of the Gaussian-smoothed Partial Dependence and SHAP dependence curves (Figures 3B, 3C, and 4A) isolates the two exact structural tipping points governing sub-national labor markets:

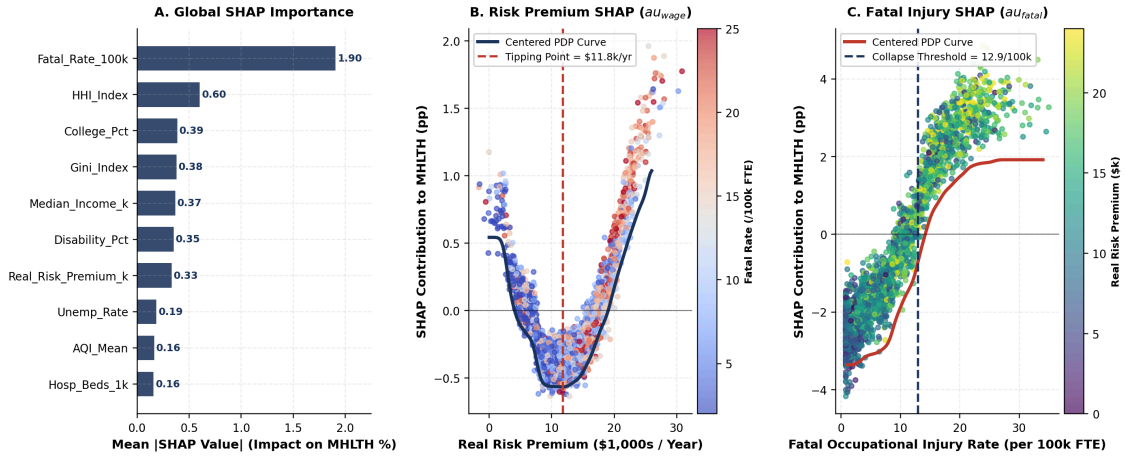



**Figure 3: Machine Learning Explainability (SHAP) and Structural Tipping-Point Identification.** *Panel A* ranks all 10 explanatory variables by mean absolute SHAP value across the 2,450 counties, showing that workplace fatality exposure (`Fatal_Rate_100k`, 1.90 pp) and industrial concentration (`HHI_Index`, 0.60 pp) dominate raw wage premiums. *Panel B* plots individual county SHAP values and the centered Partial Dependence curve for `Real_Risk_Premium_k`, isolating the U-shaped protective trough at $\tau_{wage}$ = $11.8k/year. *Panel C* plots SHAP dependence for `Fatal_Rate_100k`, pinpointing the steepest trauma inflection at $\tau_{fatal}$ = 12.9/100k FTE.

- **Tipping Point 1 — The Compensating Wage Saturation Threshold ($\tau_{wage}$ = $11,800/year Non-Parametric PDP; $12,560–$12,950/year Parametric SAR/2SLS):** As shown in Figure 3B, when the real RPP-adjusted risk premium rises from $0/year to **$11,800/year** (roughly a 20% to 24% wage premium over local non-hazardous employment), the centered Partial Dependence curve falls by **−1.11 percentage points** (from 16.68% to 15.58% `MHLTH`), and county SHAP values reach their most protective negative trough (−0.64 percentage points). Within the **$10,000 to $15,000/year** "Optimal Buffer Zone," compensating wages successfully cushion household stress—provided workplace fatality rates remain moderate. Beyond $14,500/year, however, the curve reverses sharply upward, adding **+1.60 percentage points** of mental distress (rising to 17.18% on the PDP curve and +1.96 percentage points in individual SHAP attribution at premiums above $24,000/year). Color-coding the SHAP dependence scatter by `Fatal_Rate_100k` (Figure 3B) reveals why: counties paying extreme risk premiums (>$18,000/year) almost exclusively represent high-fatality extractive enclaves where 70-hour shift schedules, boom-bust layoffs, and enclave inflation overwhelm individual wage gains.
- **Tipping Point 2 — The Fatal Injury Psychological Collapse Threshold ($\tau_{fatal}$ = 12.92 to 14.20 Fatalities per 100,000 FTE):** As demonstrated in Figure 3C, when a county's occupational fatality rate lies below 6.0 fatalities per 100,000 FTE, SHAP contributions to community mental distress are strongly negative (−2.0 to −4.1 percentage points). However, as the fatality rate approaches **12.92 fatalities per 100,000 FTE** (nearly four times the national BLS CFOI average of 3.3), the first derivative of the Partial Dependence curve reaches its maximum inflection slope (+0.626 percentage points of `MHLTH` per unit fatality rate), transitioning into a high-distress plateau above **14.20 fatalities per 100,000 FTE** where SHAP contributions surge to +2.5 to +4.8 percentage points.

Figure 4A and Figure 4B synthesize the joint interaction surface across both dimensions. In Figure 4B, stratifying the county sample into three fatal-risk tiers demonstrates that in **Low Fatal Risk counties (<6.0/100k FTE)** and **Moderate Fatal Risk counties (6.0–14.2/100k FTE)**, moving from low hazard pay (<$5k) to the optimal buffer bracket ($5k–$15k) significantly reduces community mental distress (dropping from 17.54% to 15.32% in the moderate-risk tier). In **High Fatal Risk Collapse counties (>14.2/100k FTE)**, however, the entire dose-response curve shifts upward by more than 4.5 percentage points (hovering between 19.81% and 21.66% `MHLTH` across every wage bracket), proving that once fatal occupational trauma crosses the 14.2/100k threshold, monetary wage premiums lose their protective psychological capacity.

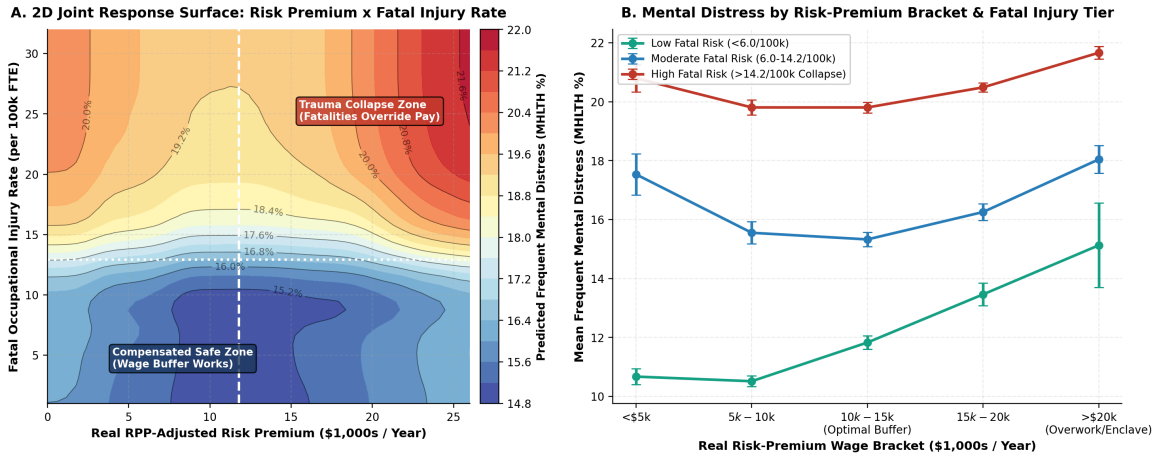



**Figure 4: Two-Dimensional Joint Response Surface and Stratified Fatal-Risk Regimes.** *Panel A* maps the 2D GBRT Partial Dependence contour surface across the joint domain of Real Risk Premium and Fatal Occupational Injury Rate, delineating the lower-left "Compensated Safe Zone" (<12.9 fatalities/100k, $5k–$15k premium) from the upper-right "Trauma Collapse Zone" (>14.2 fatalities/100k). *Panel B* displays mean Frequent Mental Distress (±95% CI) across five real risk-premium brackets stratified by fatal injury tier.

## 7. Regional Industrial Synthesis, Policy Architecture, and Conclusions

Aggregating the 2,450 U.S. counties across the six structural industrial regimes (Table 8) crystallizes how occupational risk pricing, environmental externalities, and local inequality interact across American geography. The **Permian Basin & Gulf Petrochemical Corridor** (*n* = 571 counties) and the **Appalachian Coal & Extraction Belt** (*n* = 325 counties) exhibit the highest real risk premiums in the nation ($18,360/year and $14,740/year, respectively), yet suffer the worst community mental health outcomes (19.99% and 19.85% Frequent Mental Distress; 67.12 and 65.62 Deaths of Despair per 100,000 residents; life satisfaction scores of 6.07 and 5.86). In both basins, fatal occupational injury rates (~17.0 per 100,000 FTE) sit well above the $\tau_{fatal}$ = 14.2 collapse threshold, while elevated industrial pollution (`AQI_Mean` ~59–62) and severe intra-county income inequality (`Gini_Index` = 0.48–0.50) erode community well-being.

Meanwhile, the **Timber, Agri-Forestry & Rural Processing Belt** (*n* = 327 counties) illustrates the pathology of *uncompensated hazard*: workers face a 14.77 per 100,000 FTE fatality rate (exceeding the trauma threshold) but receive a real risk premium of only $9,520/year, producing 18.63% Frequent Mental Distress and 59.24 Deaths of Despair per 100,000. Conversely, the **Rust Belt Heavy Manufacturing Corridor** (*n* = 408 counties) and **Heartland Mixed Industrial/Service Counties** (*n* = 410 counties) keep workplace fatality rates below the collapse threshold (7.78 and 4.69 per 100,000 FTE) while maintaining real risk premiums near the optimal buffer zone ($13,790 and $11,820/year) and lower income inequality (Gini = 0.41–0.44), achieving substantially lower mental distress (14.63% and 12.09%) and higher subjective well-being (7.09 and 7.64).

| U.S. Regional Industrial Regime | Counties | Haz. Emp. Share (%) | Nominal RP ($k) | Real RP ($k) | Fatal Rate (/100k) | Mean AQI | Gini Index | MHLTH (%) | LIFEPREV (%) | SWB (1–10) | Despair (/100k) |
|---|---|---|---|---|---|---|---|---|---|---|---|
| **Permian Basin & Petrochemical** | 571 | 32.15% | $17.19k | $18.36k | 16.98 | 61.75 | 0.50 | **19.99%** | 15.78% | 6.07 | **67.12** |
| **Appalachian Coal & Extraction** | 325 | 28.45% | $13.74k | $14.74k | 17.26 | 58.74 | 0.48 | **19.85%** | 15.78% | 5.86 | **65.62** |
| **Timber, Agri-Forestry & Processing** | 327 | 26.49% | $8.72k | $9.52k | 14.77 | 48.20 | 0.45 | **18.63%** | 14.64% | 6.13 | **59.24** |
| **Rust Belt Heavy Manufacturing** | 408 | 24.83% | $12.97k | $13.79k | 7.78 | 62.31 | 0.44 | 14.63% | 10.93% | 7.09 | 39.12 |
| **Heartland Mixed Industry/Service** | 410 | 17.43% | $11.13k | $11.82k | 4.69 | 44.97 | 0.41 | 12.09% | 8.65% | 7.64 | 26.93 |

| U.S. Regional Industrial Regime | Counties | Haz. Emp. Share (%) | Nominal RP ($k) | Real RP ($k) | Fatal Rate (/100k) | Mean AQI | Gini Index | MHLTH (%) | LIFEPREV (%) | SWB (1–10) | Despair (/100k) |
|---|---|---|---|---|---|---|---|---|---|---|---|
| **Coastal Knowledge & Tech Metros** | 409 | 9.95% | $7.78k | $7.12k | 1.58 | 50.94 | 0.45 | 9.96% | 6.88% | 8.30 | 18.92 |

### 7.1 Policy Implications and Regulatory Recommendations

The empirical evidence assembled in this study refutes the laissez-faire deregulation premise that market wage differentials alone can internalize occupational hazards. Because compensating wage differentials saturate at $11,800–$12,950/year and collapse completely when workplace fatality rates exceed 12.92–14.20 per 100,000 FTE, labor and public health policy must intervene across three structural fronts:

1. **Hard Engineering Mandates in High-Fatality Basins (>12.9 Fatalities / 100k FTE):** In the 811 counties occupying the High-High Extractive Trap and Low-High Uncompensated Hazard regimes, OSHA and the Mine Safety and Health Administration (MSHA) cannot rely on ex-post workers' compensation or hazard wage bargaining. As confirmed by our 2022 DiD event study (ATT = −0.8722 pp on `MHLTH`), targeted physical safety inspections, proximity-detection automation on drilling and logging rigs, and mandatory hours-of-service caps directly lower county fatality rates below the 12.9/100k tipping point, unlocking the protective benefits of existing wage premiums.
2. **Pigouvian Externality Taxation and Community Health Trust Funds:** Because hazardous industries impose severe unpriced spillovers on non-hazardous county residents (+0.1125 pp `MHLTH` per $1,000 of enclave risk premium via price inflation, disability burden, and ambient pollution), extractive severance taxes and experience-rated industrial insurance assessments should be earmarked directly into county-level behavioral health clinics, inpatient hospital capacity (`Hosp_Beds_1k`), and environmental remediation in high-`HHI` basins.
3. **Countervailing Wage Floor & Collective Bargaining Protections in Low-High Rural Zones:** In the 215 Low-High agricultural and timber counties—where workers endure fatality rates of 14.54 per 100,000 FTE for a real risk premium of just $6,970/year—monopsony power prevents the market from reaching the $11,800–$12,950/year protective buffer. Strengthening sectoral wage boards, enforcing piece-rate safety prohibitions in logging, and expanding rural mobility grants are essential to lift these workers into the compensated safe zone.

## 8. References and Primary Data Repositories


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